

Framework of n -Overarching Axioms: Overcomplete Forms and Generalized Parameterizations

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1 Introduction

This document outlines a subset of n -overarching axioms, specifically defining the concepts of “ n -forms” and “proper n -forms.” By utilizing overcomplete bases and choice functions, and by generalizing the domains of interaction, this axiomatic structure establishes a novel classificatory framework. It seamlessly accommodates both rigid coordinate systems and overlapping geometries (such as cyclic or topological covering maps), as well as internal algebraic operations and mixed-space structures.

2 The n -Form

An **n -form** is defined as a 1-overarching axiom that establishes a generalized parameterization or algebraic mapping of a set, allowing for geometric overlap and structural intersection.

Let x be an element of a primary set S . Let the input variables $y_1, y_2, \dots, y_n$ belong to sets $A_1, A_2, \dots, A_n$. These input sets are not required to be strictly disjoint from or external to S (i.e., it is permissible that $A_i = S$ for any given i , accommodating internal binary or n -ary operations). An n -form dictates that there exists a function $f : A_1 \times A_2 \times \dots \times A_n \rightarrow S$ such that:

$$x = f(y_1, y_2, \dots, y_n)$$

2.1 Properties of the n -Form

For an n -form to be valid, it must satisfy the following constraints:

- **Surjective Tuple Designation:** For every element $x \in S$, there exists **at least one** valid ordered tuple $(y_1, y_2, \dots, y_n)$ that evaluates to x . Multiple distinct tuples may map to the same x , allowing the space to be overcomplete or cyclic.

- **Variable Sharing:** The individual specific variables y_i within the tuple are not required to be uniquely bound to x .

3 The Proper n -Form

A **proper n -form** is a stricter subset of the n -form that enforces global algebraic isolation and dimensional independence, utilizing principal branches to maintain structural integrity in an overlapping space.

An n -form is considered “proper” if and only if it satisfies two additional overarching properties:

3.1 Strict Non-Uniqueness of Variables

For every variable y_i in the function f , y_i is **strictly not unique** to the output x . This ensures that every variable acts as a fully independent dimension or parameter, preventing the geometric space from collapsing into a lower dimension.

3.2 Omni-Directional Invertibility via Choice Functions

For every input variable y_i , there exists an exact function g_i that perfectly isolates y_i . Because the underlying n -form permits multiple valid tuples for a single x , the g_i functions operate as **choice functions** (defining a principal branch) to guarantee a single-valued output.

Specifically, for each variable $y_1, y_2, \dots, y_n$, there exist strictly single-valued functions $g_1, g_2, \dots, g_n$ such that:

$$\begin{aligned} y_1 &= g_1(x, y_2, \dots, y_n) \\ y_2 &= g_2(x, y_1, y_3, \dots, y_n) \\ &\dots \\ y_n &= g_n(x, y_1, y_2, \dots, y_{n-1}) \end{aligned}$$

Even if infinite valid tuples exist to satisfy the core equation $x = f(y_1, \dots, y_n)$, the function g_i is rigorously defined to select exactly one canonical value for y_i . This ensures flawless algebraic rearrangement and navigation through the space without fracturing into multi-valued branches.